

The "Severe Scrutinizer" has been summoned, and he has found his next target.

Let me begin by stating that the overarching thesis of this essay is **brilliant**. Bridging the qualitative theory of Liénard systems with the rigorous geometry of numerical integration—and explicitly diagnosing *why* symplectic methods fail in the stiff relaxation regime (i.e., the tension between A -stability and L -stability in the fast modes)—is an exceptional piece of mathematical storytelling. Your demonstration of the $O(h^{p+1})$ secular drift for RK4 versus the bounded envelope of Storer-Verlet is textbook-perfect.

However, you invited me to kill it, and I must inform you that **Section 2.1 contains two fatal algebraic and notational flaws**. You cannot publish an essay on geometric integration where the explicit algebraic expansions of your two canonical symplectic maps are wrong.

Here is the severe breakdown of what must be fixed.

1. Fatal Algebra Error: The Storer-Verlet Expansion (Eq 7)

You write that for the harmonic oscillator ($V = x^2/2$), the Storer-Verlet map evaluates to:

$$(x_1, v_2) = \left(x + hv - \frac{h^2}{2}x, \quad v - hx - \frac{h^3}{4}x + \frac{h^2}{2}v\right)$$

and you claim that direct computation gives $\det M = 1$.

This is mathematically false. The signs on your h^2 and h^3 terms in the velocity update are flipped. If you take the Jacobian M of the transformation you actually wrote down, its determinant is:

$$\det M = (1 - h^2/2)(1 + h^2/2) - h(-h - h^3/4) = 1 - h^4/4 + h^2 + h^4/4 = 1 + h^2 \neq 1$$

A map with determinant $1 + h^2$ is decidedly *not* symplectic.

The Fix: Let's derive it correctly.

1. $v_1 = v - \frac{h}{2}x$
2. $x_1 = x + hv_1 = x + hv - \frac{h^2}{2}x$ (Your x_1 is correct).
3. $v_2 = v_1 - \frac{h}{2}x_1 = \left(v - \frac{h}{2}x\right) - \frac{h}{2}\left(x + hv - \frac{h^2}{2}x\right)$
4. $v_2 = v - \frac{h}{2}x - \frac{h}{2}x - \frac{h^2}{2}v + \frac{h^3}{4}x$
5. $v_2 = v - hx + \frac{h^3}{4}x - \frac{h^2}{2}v$

Your script likely computed the determinant of the *true* Storer-Verlet map (which is 1), but you transcribed the algebraic expansion into the text with two flipped signs. Correct the text to: $v_2 = v - hx + \frac{h^3}{4}x - \frac{h^2}{2}v$.

2. Fatal Notation Clash: Implicit Midpoint (Eq 8)

You define (x_*, v_*) as the **midpoints** of the implicit midpoint rule:

"The map solves $(x_*, v_*) = (x, v) + h(v_*, -V'(x_*))/2$ in the midpoints."

You then provide explicit formulas for x_* and v_* for $V = x^2/2$:

$$x_* = \frac{-h^2x + 4hv + 4x}{h^2 + 4}, \quad v_* = \frac{-h^2v - 4hx + 4v}{h^2 + 4}$$

and then you state: "...with $\det M = 1$ and $M^T J M - J = 0$."

This is a severe conceptual collision. The formulas you wrote down are **not** the midpoints (x_*, v_*) . They are the formulas for the *next step* (x_1, v_1) . If those equations truly represented the midpoint map $(x, v) \mapsto (x_*, v_*)$, the determinant of its Jacobian would be:

$$\det M_* = \frac{4}{4 + h^2} \neq 1$$

The transformation to the *midpoint* is not volume-preserving (it shrinks phase space). It is the transformation to the *endpoint* $(x_1, v_1) = 2(x_*, v_*) - (x, v)$ that is symplectic.

The Fix: You must clarify your notation. Define (x_1, v_1) as the step update. State that solving for the midpoints yields the update map:

$$x_1 = \frac{(4 - h^2)x + 4hv}{4 + h^2}, \quad v_1 = \frac{(4 - h^2)v - 4hx}{4 + h^2}$$

(Note: I factored out your numerators, which makes the symmetry much more elegant and obvious). Then you can correctly claim that the Jacobian of the map $(x, v) \mapsto (x_1, v_1)$ has determinant 1.

3. Minor Terminology Slip: Radau IIA Order (Section 2.3)

In your table of stability functions, you list:

$$\text{"Radau IIA (order } s\text{"} \implies R_{s-1,s}(z) \text{ (Padé)}$$

This is standard numerical analysis trivia, but it is incorrect. For Radau IIA, s is the **number of stages**, not the order. An s -stage Radau IIA method has order $p = 2s - 1$. The Padé approximant $R_{s-1,s}$ corresponds to the stage count. (e.g., $s = 1$ stage is Backward Euler, which is order 1, and $R_{0,1} = 1/(1 - z)$). **The Fix:** Change "order s " to " s -stage".

The Praise (Because you earned it)

Once you fix those algebraic transcription errors, this essay is a powerhouse.

1. **The $O(h^{p+1})$ secular drift proof (Section 4.2):** Your demonstration that scaling h by a factor of 5 (from 0.05 to 0.01) drops the RK4 secular drift rate by exactly $5^5 \approx 3125$ is magnificent. It transitions geometric integration from abstract theorem to undeniable physical reality.
2. **The explanation of Splitting failure in the stiff regime (Section 6.3):** Many authors would just say "the split method fails due to stiffness." Your diagnosis—that the exact damping substep $e^{\mu(1-x^2)h/2}$ acts as an unresolvable massive *amplification* when the trajectory crosses the unstable central region $|x| < 1$ during a fast jump—is exceptionally insightful. It proves that you deeply understand the mechanics of the ODE, not just the code.

Verdict: Fix the two algebraic/notational flaws in Section 2.1 and the minor nomenclature slip in 2.3. Do that, and this is an airtight, superb companion piece.